On the end-point of Stein-Weiss inequality

Chuhan Sun and Zipeng Wang

Abstract

We give a characterization regarding power weight inequality of fractional integrals for $1 \leq p < q < \infty$. This improves the classical Stein-Weiss theorem to include the end-point $p = 1$.

1 Introduction

Let $0 < \alpha < \mathbf{N}$ and $\gamma, \delta < \mathbf{N}$. We define

$$\mathbf{I}_{\alpha\gamma\delta} f(x) \; = \; \int_{\mathbb{R}^{\mathbf{N}}} f(y) \left(\frac{1}{|x|}\right)^{\gamma} \left(\frac{1}{|x-y|}\right)^{\mathbf{N}-\alpha} \left(\frac{1}{|y|}\right)^{\delta} dy, \qquad x \neq 0. \tag{1. 1}$$

In 1928, Hardy and Littlewood [1] first investigated the $\mathbf{L}^p - \mathbf{L}^q$-regularity of $\mathbf{I}_{\alpha\gamma\delta}$ at $\mathbf{N} = 1$. Thirty years later, Stein and Weiss [3] extended this result to every higher dimensional space. Today, it bears the name of Stein-Weiss inequality.

$\diamond$ Throughout, $\mathfrak{C} > 0$ is regarded as a generic constant depending on its subindices.

Theorem A: Stein and Weiss, 1958

Let $\mathbf{I}_{\alpha\gamma\delta}$ defined in (1. 1) for $0 < \alpha < \mathbf{N}$ and $\gamma, \delta < \mathbf{N}$. We have

$$\left\|\mathbf{I}_{\alpha\gamma\delta} f\right\|_{\mathbf{L}^q(\mathbb{R}^{\mathbf{N}})} \; \leq \; \mathfrak{C}_{\alpha\,\gamma\,\delta\,p\,q} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \qquad 1 < p \leq q < \infty \tag{1. 2}$$

if

$$\gamma \; < \; \frac{\mathbf{N}}{q}, \qquad \delta \; < \; \mathbf{N}\left(\frac{p-1}{p}\right), \qquad \gamma + \delta \; \geq \; 0, \qquad \frac{\alpha}{\mathbf{N}} \; = \; \frac{1}{p} - \frac{1}{q} + \frac{\gamma+\delta}{\mathbf{N}}. \tag{1. 3}$$

Remark 1.1. *The constraints of $\alpha, \gamma, \delta, p, q$ in (1. 3) are in fact necessary. See subsection* **3.1**.

When $\gamma = \delta = 0$, **Theorem A** was proved in $\mathbb{R}^{\mathbf{N}}$ by Sobolev [2]. This is also known as Hardy-Littlewood-Sobolev inequality.

The theory of fractional integration in weighted norms has been substantially developed during the second half of 20th century. See Fefferman and Muckenhoupt [5], Muckenhoupt and Wheeden [6], Pérez [9] and Sawyer and Wheeden [7].

Our main result is an improvement of **Theorem A** to include the end-point $p = 1$.

Theorem One *Let $\mathbf{I}_{\alpha\gamma\delta}$ defined in (1. 1) for $0 < \alpha < \mathbf{N}$ and $\gamma, \delta < \mathbf{N}$. For $p > 1$, we have*

$$\left\|\mathbf{I}_{\alpha\gamma\delta} f\right\|_{\mathbf{L}^q(\mathbb{R}^\mathbf{N})} \leq \mathfrak{C}_{\alpha\ \gamma\ \delta\ p\ q} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})}, \qquad 1 < p \leq q < \infty \tag{1. 4}$$

if and only if

$$\gamma < \frac{\mathbf{N}}{q}, \qquad \delta < \mathbf{N}\left(\frac{p-1}{p}\right), \qquad \gamma + \delta \geq 0, \qquad \frac{\alpha}{\mathbf{N}} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma+\delta}{\mathbf{N}}. \tag{1. 5}$$

For $p = 1$, we have

$$\left\|\mathbf{I}_{\alpha\gamma\delta} f\right\|_{\mathbf{L}^q(\mathbb{R}^\mathbf{N})} \leq \mathfrak{C}_{\alpha\ \gamma\ \delta\ q} \left\|f\right\|_{\mathbf{L}^1(\mathbb{R}^\mathbf{N})}, \qquad 1 \leq q < \infty \tag{1. 6}$$

if and only if

$$\gamma < \frac{\mathbf{N}}{q}, \qquad \delta < 0, \qquad \gamma + \delta > 0, \qquad \frac{\alpha}{\mathbf{N}} = 1 - \frac{1}{q} + \frac{\gamma+\delta}{\mathbf{N}}. \tag{1. 7}$$

Remark 1.2. *For $p = 1$, another recent work of Nápoli and Picon [8] shows (1. 7) implying (1. 6) by using a nice 'good kernel' approximation argument.*

We prove **Theorem One** universally for $p > 1$ and $p = 1$ by improving the original framework established by Stein and Weiss [3]. This paper is organized as follows. The next section is devoted to two fundamental lemmas initially hold for $p > 1$. These results now become applicable for $p \geq 1$. We finish the proof of **Theorem One** in Section 3.

2 Two fundamental lemmas

Lemma 2.1. *Let $\Omega(u, v) \geq 0$ defined in the quadrant $\{(u, v) : u \geq 0,\ v \geq 0\}$ which is homogeneous of degree $-\mathbf{N}$ and*

$$\mathbf{A} \doteq \int_0^\infty \Omega(1, t) t^{\mathbf{N}\left(\frac{p-1}{p}\right)-1} dt < \infty. \tag{2. 1}$$

Consider

$$\mathbf{U}f(x) = \int_{\mathbb{R}^\mathbf{N}} \Omega\left(|x|, |y|\right) f(y) dy. \tag{2. 2}$$

We have

$$\left\|\mathbf{U}f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})} \leq \mathfrak{C}_\mathbf{A} \|f\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})}, \qquad 1 \leq p < \infty. \tag{2. 3}$$

Proof. Let $R = |x|$ and $r = |y|$. Write $x = R\xi$ and $y = r\eta$ of which ξ, η are unit vectors. We have

$$\mathbf{U}f(x) = \int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty \Omega(R, r) f(r\eta) r^{\mathbf{N}-1} dr d\sigma(\eta) \tag{2. 4}$$

where $d\sigma$ is the surface measure on $\mathbb{S}^{\mathbf{N}-1}$.

In particular for $\mathbf{N} = 1$, $d\sigma$ is the point measure on $\{-1, 1\}$.

Recall that Ω is homogeneous of degree $-\mathbf{N}$. Consider

$$
\begin{aligned}
&\left\{\int_0^\infty \left|\int_0^\infty \Omega(R,r) f(r\eta) r^{\mathbf{N}-1} dr\right|^p R^{\mathbf{N}-1} dR\right\}^{\frac{1}{p}} \\
&= \left\{\int_0^\infty \left|\int_0^\infty \Omega(1,t) f(tR\eta) t^{\mathbf{N}-1} dt\right|^p R^{\mathbf{N}-1} dR\right\}^{\frac{1}{p}} \\
&\leq \int_0^\infty \Omega(1,t) t^{\mathbf{N}-1} \left\{\int_0^\infty \left|f(tR\eta)\right|^p R^{\mathbf{N}-1} dR\right\}^{\frac{1}{p}} dt \qquad \textit{by Minkowski integral inequality} \qquad (2.\ 5) \\
&= \int_0^\infty \Omega(1,t) t^{\mathbf{N}\left[1-\frac{1}{p}\right]-1} \left\{\int_0^\infty \left|f(r\eta)\right|^p r^{\mathbf{N}-1} dr\right\}^{\frac{1}{p}} dt \\
&= \mathbf{A} \left\{\int_0^\infty \left|f(r\eta)\right|^p r^{\mathbf{N}-1} dr\right\}^{\frac{1}{p}}.
\end{aligned}
$$

For $\mathbf{N} \geq 2$, we find

$$
\begin{aligned}
\left\|\mathbf{U} f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})} &= \left\{\int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty \left|\int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty \Omega(R,r) f(r\eta) r^{\mathbf{N}-1} dr d\sigma(\eta)\right|^p R^{\mathbf{N}-1} dR d\sigma(\xi)\right\}^{\frac{1}{p}} \\
&= \omega_{\mathbf{N}-1}^{\frac{1}{p}} \left\{\int_0^\infty \left|\int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty \Omega(R,r) f(r\eta) r^{\mathbf{N}-1} dr d\sigma(\eta)\right|^p R^{\mathbf{N}-1} dR\right\}^{\frac{1}{p}} \\
&\leq \omega_{\mathbf{N}-1}^{\frac{1}{p}} \int_{\mathbb{S}^{\mathbf{N}-1}} \left\{\int_0^\infty \left|\int_0^\infty \Omega(R,r) f(r\eta) r^{\mathbf{N}-1} dr\right|^p R^{\mathbf{N}-1} dR\right\}^{\frac{1}{p}} d\sigma(\eta) \\
&\qquad \textit{by Minkowski integral inequality} \qquad (2.\ 6) \\
&\leq \omega_{\mathbf{N}-1}^{\frac{1}{p}} \mathbf{A} \int_{\mathbb{S}^{\mathbf{N}-1}} \left\{\int_0^\infty \left|f(r\eta)\right|^p r^{\mathbf{N}-1} dr\right\}^{\frac{1}{p}} d\sigma(\eta) \qquad \textit{by (2. 5)} \\
&\leq \mathbf{A} \omega_{\mathbf{N}-1}^{\frac{1}{p}} \left\{\int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty \left|f(r\eta)\right|^p r^{\mathbf{N}-1} dr d\sigma(\eta)\right\}^{\frac{1}{p}} \left\{\int_{\mathbb{S}^{\mathbf{N}-1}} d\sigma(\eta)\right\}^{\frac{p-1}{p}} \\
&\qquad \textit{by Hölder inequality} \\
&= \mathbf{A} \omega_{\mathbf{N}-1} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})}
\end{aligned}
$$

where

$$
\omega_{\mathbf{N}-1} = 2\pi^{\frac{\mathbf{N}}{2}} \Gamma^{-1}\left(\frac{\mathbf{N}}{2}\right)
$$

is the area of $\mathbb{S}^{\mathbf{N}-1}$.

When $\mathbf{N} = 1$, $d\sigma$ is the point measure on 1 and -1. The same estimates hold in (2. 6). $\square$

Lemma 2.2. *Define* $\Delta(t,\xi,\eta) = \left|1-2t\xi\cdot\eta+t^2\right|^{\frac{1}{2}}$ *for* $t>0$ *and* $\xi,\eta\in\mathbb{S}^{\mathbf{N}-1}$. *We have*

$$
\begin{array}{lcl}
\displaystyle\int_{\mathbb{S}^{\mathbf{N}-1}} \frac{1}{\Delta^{\mathbf{N}-\alpha}(t,\xi,\eta)}d\sigma(\xi) & = & \displaystyle\int_{\mathbb{S}^{\mathbf{N}-1}} \frac{1}{\Delta^{\mathbf{N}-\alpha}(t,\xi,\eta)}d\sigma(\eta) \\
& \leq & \mathfrak{C}\ |1-t|^{-\frac{\mathbf{N}-\alpha}{\mathbf{N}}}, \qquad t\neq 1, \qquad \xi,\eta\in\mathbb{S}^{\mathbf{N}-1}.
\end{array}
\tag{2. 7}
$$

Proof. When $\mathbf{N}=1$, $d\sigma$ is the point measure on 1 or -1. We find $\Delta(t,\xi,\eta) = |1\pm t|, t>0$ for $\xi,\eta\in\{-1,1\}$. The two integrals in (2. 7) are clearly bounded by $|1-t|^{\alpha-1}$ for $t>0, t\neq 1$.

Let $\mathbf{N}\geq 2$. Note that $\Delta(t,\xi,\eta)$ is symmetric *w.r.t* ξ and η. For $0<t<1$, we have

$$
\mathbf{P}(\xi,t\eta) \ = \ \frac{1-|t\eta|^2}{|\xi-t\eta|^{\mathbf{N}}} \ = \ \frac{1-t^2}{\Delta^{\mathbf{N}}(t,\xi,\eta)}, \qquad \xi,\eta\in\mathbb{S}^{\mathbf{N}-1} \tag{2. 8}
$$

which is the Poisson kernel on the unit sphere $\mathbb{S}^{\mathbf{N}-1}$.

A direct computation shows

$$
\Delta_\eta\mathbf{P}(\xi,t\eta) \ = \ 0, \qquad \xi\in\mathbb{S}^{\mathbf{N}-1} \tag{2. 9}
$$

where Δ_η is the Laplacian operator *w.r.t* η.

By using the mean value property of harmonic functions, we find

$$
1 \ = \ \mathbf{P}(\xi,0) \ = \ \frac{1}{\omega_{\mathbf{N}-1}}\int_{\mathbb{S}^{\mathbf{N}-1}} \mathbf{P}(\xi,t\eta)d\sigma(\eta), \qquad 0<t<1. \tag{2. 10}
$$

This further implies

$$
\frac{1}{\omega_{\mathbf{N}-1}}\int_{\mathbb{S}^{\mathbf{N}-1}} \frac{1-t^2}{\Delta^{\mathbf{N}}(t,\xi,\eta)}d\sigma(\eta) \ = \ 1, \qquad 0<t<1. \tag{2. 11}
$$

On the other hand, write $0<s=t^{-1}<1$ for $t>1$. From (2. 8), we have

$$
\begin{array}{lcl}
\displaystyle\frac{1-t^2}{\Delta^{\mathbf{N}}(t,\xi,\eta)} & = & \displaystyle\frac{1-t^2}{\left|1+2t\xi\cdot\eta+t^2\right|^{\frac{\mathbf{N}}{2}}} \ = \ \frac{t^2(s^2-1)}{t^{\mathbf{N}}\left|s^2+2s\xi\cdot\eta+1\right|^{\frac{\mathbf{N}}{2}}} \\
& = & \displaystyle -\,t^{2-\mathbf{N}}\frac{1-s^2}{\Delta^{\mathbf{N}}(s,\xi,\eta)} \ = \ -\,t^{2-\mathbf{N}}\mathbf{P}(\xi,s\eta), \qquad \xi,\eta\in\mathbb{S}^{\mathbf{N}-1}.
\end{array}
\tag{2. 12}
$$

By using (2. 10) and (2. 12), we find

$$
\begin{array}{lcl}
\displaystyle\frac{1}{\omega_{\mathbf{N}-1}}\int_{\mathbb{S}^{\mathbf{N}-1}} \frac{1-t^2}{\Delta^{\mathbf{N}}(t,\xi,\eta)}d\sigma(\eta) & = & \displaystyle -\,t^{2-\mathbf{N}}\,\frac{1}{\omega_{\mathbf{N}-1}}\int_{\mathbb{S}^{\mathbf{N}-1}} \mathbf{P}(\xi,s\eta)d\sigma(\eta) \\
& = & -\,t^{2-\mathbf{N}}, \qquad t>1.
\end{array}
\tag{2. 13}
$$

By putting together (2. 11) and (2. 13), we obtain

$$
\frac{1}{\omega_{\mathbf{N}-1}}\int_{\mathbb{S}^{\mathbf{N}-1}} \frac{1}{\Delta^{\mathbf{N}}(t,\xi,\eta)}d\sigma(\eta) \ \leq \ \frac{1}{|1-t^2|} \ < \ \frac{1}{|1-t|}. \tag{2. 14}
$$

Lastly, by applying Hölder's inequality, we have

$$
\begin{aligned}
\int_{\mathbb{S}^{\mathbf{N}-1}} \frac{1}{\Delta^{\mathbf{N}-\alpha}(t,\xi,\eta)} d\sigma(\eta) &\le \left\{\int_{\mathbb{S}^{\mathbf{N}-1}} \left[\frac{1}{\Delta^{\mathbf{N}-\alpha}(t,\xi,\eta)}\right]^{\frac{\mathbf{N}}{\mathbf{N}-\alpha}} d\sigma(\eta)\right\}^{\frac{\mathbf{N}-\alpha}{\mathbf{N}}} \left\{\int_{\mathbb{S}^{\mathbf{N}-1}} d\sigma(\eta)\right\}^{\frac{\alpha}{\mathbf{N}}} \\
&= \left\{\int_{\mathbb{S}^{\mathbf{N}-1}} \frac{1}{\Delta^{\mathbf{N}}(t,\xi,\eta)} d\sigma(\eta)\right\}^{\frac{\mathbf{N}-\alpha}{\mathbf{N}}} (\omega_{\mathbf{N}-1})^{\frac{\alpha}{\mathbf{N}}} \qquad (2.\ 15) \\
&\le \left(\frac{1}{|1-t|}\right)^{\frac{\mathbf{N}-\alpha}{\mathbf{N}}} (\omega_{\mathbf{N}-1})^{\frac{\mathbf{N}-\alpha}{\mathbf{N}}} (\omega_{\mathbf{N}-1})^{\frac{\alpha}{\mathbf{N}}} \qquad \textit{by (2. 14)} \\
&= \omega_{\mathbf{N}-1}\, |1-t|^{-\frac{\mathbf{N}-\alpha}{\mathbf{N}}}.
\end{aligned}
$$

3 Proof of Theorem One

3.1 Necessity of the constraints in (1. 5) and (1. 7)

Case 1 Let $p > 1$. Consider $f(x) = \chi_Q(x)|x|^{-\delta\left(\frac{p}{p-1}\right)}$. As a well known result, (1. 4) implies

$$
\sup_{Q\subset\mathbb{R}^{\mathbf{N}}} |Q|^{\frac{\alpha}{\mathbf{N}}-\frac{1}{p}+\frac{1}{q}} \left\{\frac{1}{|Q|}\int_Q \left(\frac{1}{|x|}\right)^{\gamma q} dx\right\}^{\frac{1}{q}} \left\{\frac{1}{|Q|}\int_Q \left(\frac{1}{|x|}\right)^{\delta\left(\frac{p}{p-1}\right)} dx\right\}^{\frac{p-1}{p}} < \infty. \qquad (3.\ 1)
$$

We essentially need $\gamma < \mathbf{N}/q$ and $\delta < \mathbf{N}\left(\frac{p-1}{p}\right)$ for which $|x|^{-\gamma q}$ and $|x|^{-\delta\left(\frac{p}{p-1}\right)}$ are locally integrable. By changing dilations inside (3. 1), we find $\frac{\alpha}{\mathbf{N}} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma+\delta}{\mathbf{N}}$ is an necessity. Moreover, we claim $\frac{\alpha}{\mathbf{N}} \ge \frac{1}{p} - \frac{1}{q}$. Together with $\frac{\alpha}{\mathbf{N}} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma+\delta}{\mathbf{N}}$, we must have $\gamma + \delta \ge 0$.

Suppose $\frac{\alpha}{\mathbf{N}} < \frac{1}{p} - \frac{1}{q}$. Let Q centered on some $x_o \ne 0$. By shrinking Q to x_o and applying Lebesgue's Differentiation Theorem, we find

$$
\left\{\frac{1}{|Q|}\int_Q \left(\frac{1}{|x|}\right)^{\gamma q} dx\right\}^{\frac{1}{q}} \left\{\frac{1}{|Q|}\int_Q \left(\frac{1}{|x|}\right)^{\delta\left(\frac{p}{p-1}\right)} dx\right\}^{\frac{p-1}{p}} = |x_o|^{-(\gamma+\delta)} > 0. \qquad (3.\ 2)
$$

On the other hand, $|Q|^{\frac{\alpha}{\mathbf{N}}-\frac{1}{p}+\frac{1}{q}} \longrightarrow \infty$. We reach a contradiction to (3. 1).

Case 2 Let $p = 1$. Denote Q as any cube in $\mathbb{R}^{\mathbf{N}}$. Choose $f = \chi_Q$ which is an indicator function supported in Q. The $\mathbf{L}^p \longrightarrow \mathbf{L}^q$-norm inequality in (1. 6) implies

$$
\sup_{Q\subset\mathbb{R}^{\mathbf{N}}} |Q|^{\frac{\alpha}{\mathbf{N}}-1+\frac{1}{q}} \left\{\frac{1}{|Q|}\int_Q \left(\frac{1}{|x|}\right)^{\gamma q} dx\right\}^{\frac{1}{q}} \left\{\frac{1}{|Q|}\int_Q \left(\frac{1}{|x|}\right)^{\delta} dx\right\} < \infty. \qquad (3.\ 3)
$$

A standard exercise of changing dilations inside (3. 3) shows that $\frac{\alpha}{\mathbf{N}} = 1 - \frac{1}{q} + \frac{\gamma+\delta}{\mathbf{N}}$ is an necessary condition. Moreover, it is essential to have $\gamma < \frac{\mathbf{N}}{q}$ for the local integrability of $|x|^{-\gamma q}$. We claim $\frac{\alpha}{\mathbf{N}} - 1 + \frac{1}{q} \ge 0$. Together with $\frac{\alpha}{\mathbf{N}} = 1 - \frac{1}{q} + \frac{\gamma+\delta}{\mathbf{N}}$, we must have $\gamma + \delta \ge 0$.

Suppose $\frac{\alpha}{\mathbf{N}} - 1 + \frac{1}{q} < 0$. Let Q centered on some $x_o \neq 0$. By shrinking Q to x_o and applying Lebesgue's Differentiation Theorem, we find

$$\left\{ \frac{1}{|Q|} \int_Q \left(\frac{1}{|x|} \right)^{\gamma q} dx \right\}^{\frac{1}{q}} \left\{ \frac{1}{|Q|} \int_Q \left(\frac{1}{|x|} \right)^{\delta} dx \right\} \; = \; |x_o|^{-(\gamma+\delta)} \; > \; 0. \tag{3. 4}$$

On the other hand, $|Q|^{\frac{\alpha}{\mathbf{N}} - 1 + \frac{1}{q}} \longrightarrow \infty$. This contradicts to (3. 3).

Furthermore, at $p = 1$, we essentially need $\gamma + \delta > 0$. This can be observed by carrying out the simple counter-example below.

Recall $\mathbf{I}_{\alpha\gamma\delta} f$ defined in (1. 1). Suppose $\gamma + \delta = 0$. The homogeneity condition in (1. 7) becomes $\frac{\alpha}{\mathbf{N}} = 1 - \frac{1}{q}$. Choose $\{f_n\}_{n \geq 1}$ to be a family of good kernels whose limit is a Dirac delta centered on some $x_o \neq 0$. Consequently, we find

$$\begin{aligned} \mathbf{I}_{\alpha\gamma\delta} f_n(x) \; &= \; \int_{\mathbb{R}^{\mathbf{N}}} f_n(y) \left(\frac{1}{|x|} \right)^{\gamma} \left(\frac{1}{|x-y|} \right)^{\mathbf{N}-\alpha} \left(\frac{1}{|y|} \right)^{\delta} dy \\ &= \; \left(\frac{1}{|x|} \right)^{\gamma} \left(\frac{1}{|x-x_o|} \right)^{\mathbf{N}-\alpha} \left(\frac{1}{|x_o|} \right)^{\delta}, \qquad \text{as } n \longrightarrow \infty \end{aligned} \tag{3. 5}$$

whenever $x \neq x_o$. Because $q[\mathbf{N} - \alpha] = \mathbf{N}$, the q-th power of the function in (3. 5) is locally non-integrable in a neighborhood of x_o.

Let $f \geq 0$, $f \in \mathbf{L}^1(\mathbb{R}^{\mathbf{N}})$. Moreover, f is supported in the unit ball, denoted by $\mathbf{B}$. We have

$$\begin{aligned} \mathbf{I}_{\alpha\gamma\delta} f(x) \; &= \; \int_{\mathbb{R}^{\mathbf{N}}} f(y) \left(\frac{1}{|x|} \right)^{\gamma} \left(\frac{1}{|x-y|} \right)^{\mathbf{N}-\alpha} \left(\frac{1}{|y|} \right)^{\delta} dy \\ &\geq \; \chi(|x| > 10) \int_{\mathbf{B}} f(y) \left(\frac{1}{|x|} \right)^{\gamma} \left(\frac{1}{|x-y|} \right)^{\mathbf{N}-\alpha} \left(\frac{1}{|y|} \right)^{\delta} dy \\ &> \; 2^{\alpha-\mathbf{N}} \left(\frac{1}{|x|} \right)^{\mathbf{N}-\alpha+\gamma} \chi(|x| > 10) \int_{\mathbf{B}} f(y) \left(\frac{1}{|y|} \right)^{\delta} dy. \end{aligned} \tag{3. 6}$$

Observe that if $\delta \geq 0$, then $\frac{\alpha}{\mathbf{N}} = 1 - \frac{1}{q} + \frac{\gamma+\delta}{\mathbf{N}}$ implies $\mathbf{N} - \alpha + \gamma \leq \frac{\mathbf{N}}{q}$. Consequently, $[\mathbf{I}_{\alpha\gamma\delta} f]^q$ cannot be integrable in $\mathbb{R}^{\mathbf{N}}$. Therefore, we also need $\delta < 0$.

3.2 The $\mathbf{L}^p \longrightarrow \mathbf{L}^q$-norm inequality in (1. 4) and (1. 6)

Consider

$$\mathbf{I}_{\alpha\gamma\delta} f(x) \; = \; \mathbf{U}_1 f(x) + \mathbf{U}_2 f(x) + \mathbf{U}_3 f(x), \qquad f \geq 0 \tag{3. 7}$$

where

$$\begin{aligned} \mathbf{U}_1 f(x) \; &= \; \int_{\mathbb{R}^{\mathbf{N}}} f(y) \Omega_1(x,y) dy, \\ \Omega_1(x,y) \; &= \; \begin{cases} \left(\frac{1}{|x|} \right)^{\gamma} \left(\frac{1}{|x-y|} \right)^{\mathbf{N}-\alpha} \left(\frac{1}{|y|} \right)^{\delta}, & 0 \leq \frac{|y|}{|x|} \leq \frac{1}{2}, \\ 0, & \frac{|y|}{|x|} > \frac{1}{2}; \end{cases} \end{aligned} \tag{3. 8}$$

$$\mathbf{U}_2 f(x) \;=\; \int_{\mathbb{R}^{\mathbf{N}}} f(y)\Omega_2(x,y)dy,$$

$$\Omega_2(x,y) \;=\; \begin{cases} \left(\frac{1}{|x|}\right)^{\gamma}\left(\frac{1}{|x-y|}\right)^{\mathbf{N}-\alpha}\left(\frac{1}{|y|}\right)^{\delta}, & \frac{|y|}{|x|}\geq 2, \\ 0, & 0\leq\frac{|y|}{|x|}<2 \end{cases} \tag{3. 9}$$

and

$$\mathbf{U}_3 f(x) \;=\; \int_{\mathbb{R}^{\mathbf{N}}} f(y)\Omega_3(x,y)dy,$$

$$\Omega_3(x,y) \;=\; \begin{cases} \left(\frac{1}{|x|}\right)^{\gamma}\left(\frac{1}{|x-y|}\right)^{\mathbf{N}-\alpha}\left(\frac{1}{|y|}\right)^{\delta}, & \frac{1}{2}<\frac{|y|}{|x|}<2, \\ 0, & 0\leq\frac{|y|}{|x|}\leq\frac{1}{2} \text{ or } \frac{|y|}{|x|}\geq\frac{1}{2}. \end{cases} \tag{3. 10}$$

We aim to show that each $\mathbf{U}_i f, i = 1,2,3$ satisfies the $\mathbf{L}^p \longrightarrow \mathbf{L}^q$-norm inequality in (1. 6). The first two can be obtained by refining the proof of Stein and Weiss [3] and using the two lemmas from the previous section. The crucial part of our estimates comes when dealing with $\mathbf{U}_3 f$ for which we go through an interpolation argument of changing measures.

Case One Let $1 \leq p = q < \infty$. The homogeneity condition $\frac{\alpha}{\mathbf{N}} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma+\delta}{\mathbf{N}}$ implies $\alpha = \gamma + \delta$.

Note that $|x-y| \geq \frac{1}{2}|x|$ if $\frac{|y|}{|x|} \leq \frac{1}{2}$. From (3. 8), we find

$$\Omega_1(x,y) \;\leq\; 2^{\mathbf{N}-\alpha}\begin{cases} \left(\frac{1}{|x|}\right)^{\mathbf{N}-\alpha+\gamma}\left(\frac{1}{|y|}\right)^{\delta}, & 0\leq\frac{|y|}{|x|}\leq\frac{1}{2}, \\ 0, & \frac{|y|}{|x|}>\frac{1}{2}. \end{cases} \tag{3. 11}$$

Because $\delta < \mathbf{N}\left(\frac{p-1}{p}\right)$, we have

$$\begin{aligned} \mathbf{A}_1 \;&\doteq\; \int_0^\infty \Omega_1(1,t)t^{\mathbf{N}\left(\frac{p-1}{p}\right)-1}dt \\ &\leq\; 2^{\mathbf{N}-\alpha}\int_0^{\frac{1}{2}} t^{\mathbf{N}\left(\frac{p-1}{p}\right)-\delta-1}dt \;<\; \infty. \end{aligned} \tag{3. 12}$$

By applying Lemma 2.1, we obtain

$$\left\|\mathbf{U}_1 f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \;\leq\; \mathfrak{C}_{\alpha\,\delta\,p}\,\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \qquad 1\leq p<\infty. \tag{3. 13}$$

On the other hand, $|x-y| \geq \frac{1}{2}|y|$ if $\frac{|y|}{|x|} \geq 2$. From (3. 9), we find

$$\Omega_2(x,y) \;\leq\; 2^{\mathbf{N}-\alpha}\begin{cases} \left(\frac{1}{|x|}\right)^{\gamma}\left(\frac{1}{|y|}\right)^{\mathbf{N}-\alpha+\delta}, & \frac{|y|}{|x|}\geq 2, \\ 0, & 0\leq\frac{|y|}{|x|}<2. \end{cases} \tag{3. 14}$$

Because $\gamma < \mathbf{N}/q = \mathbf{N}/p$ and $\alpha = \gamma + \delta$, we have

$$
\begin{aligned}
\mathbf{A}_2 \ &\doteq\ \int_0^\infty \Omega_2(1,t)t^{\mathbf{N}\left(\frac{p-1}{p}\right)-1}dt \\
&\leq\ 2^{\mathbf{N}-\alpha}\int_2^\infty t^{\mathbf{N}\left(\frac{p-1}{p}\right)-\mathbf{N}+\alpha-\delta-1}dt \ =\ 2^{\mathbf{N}-\alpha}\int_2^\infty t^{-\frac{\mathbf{N}}{p}+\gamma-1}dt \ <\ \infty.
\end{aligned}
\tag{3. 15}
$$

By applying Lemma 2.1, we obtain

$$
\left\|\mathbf{U}_2 f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})}\ \leq\ \mathfrak{C}_{\alpha\ \gamma\ p}\ \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})}, \qquad 1\leq p<\infty. \tag{3. 16}
$$

Write $x = R\xi$ and $y = r\eta$ for $\xi, \eta \in \mathbb{S}^{\mathbf{N}-1}$. Recall Ω_3 defined in (3. 10) which is homogeneous of degree $-\mathbf{N}$.

We have

$$
\begin{aligned}
\mathbf{U}_3 f(x) \ &=\ \int_{\mathbb{R}^\mathbf{N}} \Omega_3(x,y)f(y)dy \\
&=\ \int_{\mathbb{S}^{\mathbf{N}-1}}\int_0^\infty \Omega_3(R\xi, r\eta)f(r\eta)r^{\mathbf{N}-1}drd\sigma(\eta) \\
&=\ \int_{\mathbb{S}^{\mathbf{N}-1}}\int_0^\infty \Omega_3(\xi, t\eta)f(tR\eta)t^{\mathbf{N}-1}dtd\sigma(\eta) \\
&=\ \int_{\mathbb{S}^{\mathbf{N}-1}}\int_{\frac{1}{2}}^2 \frac{1}{|\xi-t\eta|^{\mathbf{N}-\alpha}}f(tR\eta)t^{\mathbf{N}-1-\delta}dtd\sigma(\eta) \qquad \textit{by (3. 10)} \\
&=\ \int_{\mathbb{S}^{\mathbf{N}-1}}\int_{\frac{1}{2}}^2 \frac{1}{\left[(\xi-t\eta)\cdot(\xi-t\eta)\right]^{\frac{\mathbf{N}-\alpha}{2}}}f(tR\eta)t^{\mathbf{N}-1-\delta}dtd\sigma(\eta) \\
&=\ \int_{\mathbb{S}^{\mathbf{N}-1}}\int_{\frac{1}{2}}^2 \frac{1}{|1-2t\xi\cdot\eta+t^2|^{\frac{\mathbf{N}-\alpha}{2}}}f(tR\eta)t^{\mathbf{N}-1-\delta}dtd\sigma(\eta) \\
&\lesssim\ \int_{\mathbb{S}^{\mathbf{N}-1}}\int_{\frac{1}{2}}^2 \frac{1}{\Delta^{\mathbf{N}-\alpha}(t,\xi,\eta)}f(tR\eta)dtd\sigma(\eta).
\end{aligned}
\tag{3. 17}
$$

In particular, for $\mathbf{N} = 1$, $d\sigma$ is a point measure on 1 and -1 inside (3. 17). We find

$$
\mathbf{U}_3 f(x)\ \lesssim\ \int_{\frac{1}{2}}^2 |1-t|^{\alpha-1}\Big[f(tR)+f(-tR)\Big]dt. \tag{3. 18}
$$

From (3. 17), we have

$$
\begin{aligned}
\left\|\mathbf{U}_3 f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \lesssim& \left\{\int_{\mathbb{S}^{\mathbf{N}-1}}\int_0^\infty \left\{\int_{\mathbb{S}^{\mathbf{N}-1}}\int_{\frac{1}{2}}^2 \frac{1}{\Delta^{\mathbf{N}-\alpha}(t,\xi,\eta)} f(tR\eta)dtd\sigma(\eta)\right\}^p R^{\mathbf{N}-1}dRd\sigma(\xi)\right\}^{\frac{1}{p}}\\
\lesssim& \int_{\frac{1}{2}}^2 \left\{\int_0^\infty \int_{\mathbb{S}^{\mathbf{N}-1}} \left\{\int_{\mathbb{S}^{\mathbf{N}-1}} \frac{1}{\Delta^{\mathbf{N}-\alpha}(t,\xi,\eta)} f(tR\eta)d\sigma(\eta)\right\}^p d\sigma(\xi)R^{\mathbf{N}-1}dR\right\}^{\frac{1}{p}} dt\\
&\qquad \textit{by Minkowski integral inequality}\\
\lesssim& \int_{\frac{1}{2}}^2 \left\{\int_0^\infty \int_{\mathbb{S}^{\mathbf{N}-1}} \left\{\int_{\mathbb{S}^{\mathbf{N}-1}} \frac{[f(tR\eta)]^p}{\Delta^{\mathbf{N}-\alpha}(t,\xi,\eta)} d\sigma(\eta)\right\}\left\{\int_{\mathbb{S}^{\mathbf{N}-1}} \frac{1}{\Delta^{\mathbf{N}-\alpha}(t,\xi,\eta)} d\sigma(\eta)\right\}^{p-1} d\sigma(\xi)R^{\mathbf{N}-1}dR\right\}^{\frac{1}{p}} dt\\
&\qquad \textit{by Hölder inequality}\\
\lesssim& \int_{\frac{1}{2}}^2 \left\{\int_0^\infty \int_{\mathbb{S}^{\mathbf{N}-1}} \left\{\int_{\mathbb{S}^{\mathbf{N}-1}} \frac{[f(tR\eta)]^p}{\Delta^{\mathbf{N}-\alpha}(t,\xi,\eta)} d\sigma(\eta)\right\}|1-t|^{\left[\frac{\alpha-\mathbf{N}}{\mathbf{N}}\right](p-1)} d\sigma(\xi)R^{\mathbf{N}-1}dR\right\}^{\frac{1}{p}} dt\\
&\qquad \textit{by Lemma 2.2}\\
=& \int_{\frac{1}{2}}^2 \left\{\int_0^\infty \int_{\mathbb{S}^{\mathbf{N}-1}} \left\{|1-t|^{\left[\frac{\alpha-\mathbf{N}}{\mathbf{N}}\right](p-1)} \int_{\mathbb{S}^{\mathbf{N}-1}} \frac{1}{\Delta^{\mathbf{N}-\alpha}(t,\xi,\eta)} d\sigma(\xi)\right\}[f(tR\eta)]^p R^{\mathbf{N}-1}dRd\sigma(\eta)\right\}^{\frac{1}{p}} dt\\
\lesssim& \int_{\frac{1}{2}}^2 \left\{|1-t|^{\left[\frac{\alpha-\mathbf{N}}{\mathbf{N}}\right]p} \int_0^\infty \int_{\mathbb{S}^{\mathbf{N}-1}} [f(tR\eta)]^p R^{\mathbf{N}-1}dRd\sigma(\eta)\right\}^{\frac{1}{p}} dt \qquad \textit{by Lemma 2.2}\\
=& \int_{\frac{1}{2}}^2 \left\{\int_0^\infty \int_{\mathbb{S}^{\mathbf{N}-1}} [f(tR\eta)]^p d\sigma(\eta)R^{\mathbf{N}-1}dR\right\}^{\frac{1}{p}} |1-t|^{\frac{\alpha-\mathbf{N}}{\mathbf{N}}} dt\\
=& \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \int_{\frac{1}{2}}^2 |1-t|^{\frac{\alpha-\mathbf{N}}{\mathbf{N}}} dt \leq \mathfrak{C}_\alpha \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \qquad 1\leq p<\infty.
\end{aligned}
\tag{3. 19}
$$

Moreover, by using (3. 18), we find

$$
\begin{aligned}
\left\|\mathbf{U}_3 f\right\|_{\mathbf{L}^p(\mathbb{R})} \lesssim& \left\{\int_0^\infty \left\{\int_{\frac{1}{2}}^2 |1-t|^{\alpha-1}\Big[f(tR)+f(-tR)\Big]dt\right\}^p dR\right\}^{\frac{1}{p}}\\
\leq& \int_{\frac{1}{2}}^2 \left\{\int_0^\infty \Big[f(tR)+f(-tR)\Big]^p dR\right\}^p |1-t|^{\alpha-1}dt\\
&\qquad \textit{by Minkowski integral inequality}\\
=& \left\|f\right\|_{\mathbf{L}^p(\mathbb{R})} \int_{\frac{1}{2}}^2 |1-t|^{\alpha-1}dt\\
\leq& \mathfrak{C}_\alpha \left\|f\right\|_{\mathbf{L}^p(\mathbb{R})}, \qquad 1\leq p<\infty.
\end{aligned}
\tag{3. 20}
$$

Case Two Consider $1 \leq p < q < \infty$. Assert

$$\mathbf{V}_\delta f(x) \;=\; |x|^{-\mathbf{N}+\delta} \int_{|y|<|x|} |y|^{-\delta} f(y)dy, \qquad \delta < \mathbf{N}\left(\tfrac{p-1}{p}\right). \tag{3. 21}$$

We claim

$$\left\|\mathbf{V}_\delta f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})} \;\leq\; \mathfrak{C}_{\delta\, p}\, \|f\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})}, \qquad 1 \leq p < \infty. \tag{3. 22}$$

Write

$$\mathbf{V}_\delta f(x) \;=\; \int_{\mathbb{R}^n} \Omega(|x|,|y|) f(y)dy, \qquad \Omega(u,v) \;=\; \begin{cases} u^{-\mathbf{N}+\delta} v^{-\delta} & \text{if } \; v<u \\ 0 & \text{otherwise.} \end{cases} \tag{3. 23}$$

Observe that Ω in (3. 23) is homogeneous of degree $-\mathbf{N}$. Moreover,

$$\int_0^\infty \Omega(1,t) t^{\mathbf{N}\left(\frac{p-1}{p}\right)-1} dt \;=\; \int_0^1 t^{\mathbf{N}\left(\frac{p-1}{p}\right)-\delta-1} dt \;<\; \infty \tag{3. 24}$$

provided by $\delta < \mathbf{N}\left(\frac{p-1}{p}\right)$. Lemma 2.1 implies (3. 22).

On the other hand, $\mathbf{V}_\delta f$ defined in (3. 21) satisfies

$$\begin{aligned} \mathbf{V}_\delta f(x) \;&\leq\; |x|^{-\mathbf{N}+\delta} \left\{ \int_{|y|<|x|} |y|^{-\delta\left(\frac{p}{p-1}\right)} dy \right\}^{\frac{p-1}{p}} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})} \qquad \textit{by Hölder inequality} \\ &\leq\; \mathfrak{C}_{\delta\, p}\, |x|^{-\frac{\mathbf{N}}{p}} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^\mathbf{N})}, \qquad 1 \leq p < \infty. \end{aligned} \tag{3. 25}$$

Recall $\mathbf{U}_1 f$ defined in (3. 8). Note that $0 \leq \frac{|y|}{|x|} \leq \frac{1}{2}$ implies $\frac{1}{2}|x| \leq |x| - |y| \leq |x-y|$. Let $f, g \geq 0$ and $f \in \mathbf{L}^p(\mathbb{R}^\mathbf{N})$, $g \in \mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^\mathbf{N})$. We have

$$\begin{aligned} \int_{\mathbb{R}^\mathbf{N}} \mathbf{U}_1 f(x) g(x) dx \;&=\; \int_{\mathbb{R}^\mathbf{N}} \left\{ \int_{|y|\leq\frac{1}{2}|x|} \frac{f(y)g(x)}{|x|^\gamma\; |x-y|^{\mathbf{N}-\alpha}\; |y|^\delta} dy \right\} dx \\ &\lesssim\; \int_{\mathbb{R}^\mathbf{N}} \left\{ \int_{|y|<|x|} \frac{f(y)g(x)}{|x|^{\gamma+\mathbf{N}-\alpha}\; |y|^\delta} dy \right\} dx \\ &=\; \int_{\mathbb{R}^\mathbf{N}} |x|^{\alpha-(\gamma+\delta)} g(x) \left\{ |x|^{-\mathbf{N}+\delta} \int_{|y|<|x|} f(y)|y|^{-\delta} dy \right\} dx \\ &=\; \int_{\mathbb{R}^\mathbf{N}} |x|^{\alpha-(\gamma+\delta)} g(x) \mathbf{V}_\delta f(x) dx \\ &\leq\; \left\{ \int_{\mathbb{R}^\mathbf{N}} |x|^{\left[\alpha-(\gamma+\delta)\right]q} \left(\mathbf{V}_\delta f\right)^q (x) dx \right\}^{\frac{1}{q}} \left\|g\right\|_{\mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^\mathbf{N})} \quad \textit{by Hölder inequality.} \end{aligned} \tag{3. 26}$$

Let $\frac{\alpha}{\mathbf{N}}=\frac{1}{p}-\frac{1}{q}+\frac{\gamma+\delta}{\mathbf{N}}, 1\le p<q<\infty$. We find

$$
\begin{aligned}
&\left\{\int_{\mathbb{R}^{\mathbf{N}}}|x|^{\left[\alpha-(\gamma+\delta)\right]q}\Big(\mathbf{V}_\delta f\Big)^q(x)dx\right\}^{\frac{1}{q}}\\
\le\;&\left\{\int_{\mathbb{R}^{\mathbf{N}}}|x|^{\left[\alpha-(\gamma+\delta)\right]q}|x|^{-\mathbf{N}\left[\frac{q}{p}-1\right]}\left\|f\right\|^{q-p}_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}\Big(\mathbf{V}_\delta f\Big)^p(x)dx\right\}^{\frac{1}{q}} \qquad \textit{by (3. 25)}\\
=\;&\left\|f\right\|^{1-\frac{p}{q}}_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}\left\{\int_{\mathbb{R}^{\mathbf{N}}}\Big(\mathbf{V}_\delta f\Big)^p(x)dx\right\}^{\frac{1}{q}}\\
\le\;&\mathfrak{C}_{\delta\,p}\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \qquad \textit{by (3. 22)}.
\end{aligned}
\tag{3. 27}
$$

From (3. 26)-(3. 27), we conclude

$$\left\|\mathbf{U}_1 f\right\|_{\mathbf{L}^q(\mathbb{R}^{\mathbf{N}})}\;\le\;\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \qquad 1\le p<q<\infty. \tag{3. 28}$$

Consider

$$\mathbf{V}_\gamma g(x)\;=\;|x|^{-\mathbf{N}+\gamma}\int_{|y|<|x|}|y|^{-\gamma}g(y)dy, \qquad \gamma<\frac{\mathbf{N}}{q}. \tag{3. 29}$$

We claim

$$\left\|\mathbf{V}_\gamma g\right\|_{\mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^{\mathbf{N}})}\;\le\;\mathfrak{C}_{\gamma\,q}\,\|g\|_{\mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^{\mathbf{N}})}, \qquad 1<q<\infty. \tag{3. 30}$$

Write

$$\mathbf{V}_\gamma g(x)\;=\;\int_{\mathbb{R}^n}\Omega(|x|,|y|)g(y)dy, \qquad \Omega(u,v)\;=\;\begin{cases} u^{-\mathbf{N}+\gamma}v^{-\gamma} & \text{if } v<u\\ 0 & \text{otherwise.}\end{cases} \tag{3. 31}$$

Observe that Ω in (3. 31) is homogeneous of degree $-\mathbf{N}$. Moreover,

$$\int_0^\infty\Omega(1,t)t^{\frac{\mathbf{N}}{q}-1}dt\;=\;\int_0^1 t^{\frac{\mathbf{N}}{q}-\gamma-1}dt\;<\;\infty \tag{3. 32}$$

provided by $\gamma<\frac{\mathbf{N}}{q}$. Lemma 2.1 implies (3. 30).

On the other hand, $\mathbf{V}_\gamma g$ defined in (3. 29) satisfies

$$
\begin{aligned}
\mathbf{V}_\gamma g(y)\;&=\;|y|^{-\mathbf{N}+\gamma}\int_{|x|<|y|}|x|^{-\gamma}g(x)dx\\
&\le\;|y|^{-\mathbf{N}+\gamma}\left\{\int_{|x|<|y|}|x|^{-\gamma q}dx\right\}^{\frac{1}{q}}\left\|g\right\|_{\mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^{\mathbf{N}})} \qquad \textit{by Hölder inequality}\\
&\le\;\mathfrak{C}_{\gamma\,q}\,|y|^{-\mathbf{N}\left(\frac{q-1}{q}\right)}\left\|g\right\|_{\mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^{\mathbf{N}})}.
\end{aligned}
\tag{3. 33}
$$

Recall $\mathbf{U}_2 f$ defined in (3. 9). Note that $\frac{|y|}{|x|} \geq 2$ implies $\frac{1}{2}|y| \leq |y| - |x| \leq |x-y|$. We have

$$\begin{aligned}\int_{\mathbb{R}^{\mathbf{N}}} \mathbf{U}_2 f(x) g(x) dx &= \int_{\mathbb{R}^{\mathbf{N}}} \left\{ \int_{|y|\geq 2|x|} \frac{f(y)}{|x|^{\gamma}\, |x-y|^{\mathbf{N}-\alpha} |y|^{\delta}} dy \right\} g(x) dx \\ &\lesssim \int_{\mathbb{R}^{\mathbf{N}}} \left\{ \int_{|y|>|x|} \frac{f(y)g(x)}{|x|^{\gamma}\, |y|^{\mathbf{N}-\alpha+\delta}} dy \right\} dx.\end{aligned} \tag{3. 34}$$

By using (3. 34) and Tonelli's theorem, we have

$$\begin{aligned}\int_{\mathbb{R}^{\mathbf{N}}} \mathbf{U}_2 f(x) g(x) dx &\lesssim \int_{\mathbb{R}^{\mathbf{N}}} |y|^{\alpha-(\gamma+\delta)} f(y) \left\{ |y|^{-\mathbf{N}+\gamma} \int_{|x|<|y|} g(x) |x|^{-\gamma} dx \right\} dy \\ &= \int_{\mathbb{R}^{\mathbf{N}}} |y|^{\alpha-(\gamma+\delta)} f(y) \mathbf{V}_{\gamma} g(y) dy \\ &\leq \left\| f \right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \left\{ \int_{\mathbb{R}^{\mathbf{N}}} |y|^{\left[\alpha-(\gamma+\delta)\right]\frac{p}{p-1}} \left(\mathbf{V}_{\gamma} g\right)^{\frac{p}{p-1}}(y) dy \right\}^{\frac{p-1}{p}} \qquad \textit{by Hölder inequality.}\end{aligned} \tag{3. 35}$$

Let $\frac{\alpha}{\mathbf{N}} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma+\delta}{\mathbf{N}}, 1 \leq p < q < \infty$. For $p = 1$, we find

$$\begin{aligned}\left\| |y|^{\alpha-(\gamma+\delta)} \mathbf{V}_{\gamma} g(y) \right\|_{\mathbf{L}^{\infty}(\mathbb{R}^{\mathbf{N}})} &\leq \mathfrak{C}_{\gamma\ q}\, |y|^{\alpha-(\gamma+\delta)} |y|^{-\mathbf{N}\left(\frac{q-1}{q}\right)} \left\| g \right\|_{\mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^{\mathbf{N}})} \qquad \text{by (3. 33)} \\ &= \mathfrak{C}_{\gamma\ q} \left\| g \right\|_{\mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^{\mathbf{N}})}.\end{aligned} \tag{3. 36}$$

For $p > 1$, write

$$\begin{aligned}&\left\{ \int_{\mathbb{R}^{\mathbf{N}}} |y|^{\left[\alpha-(\gamma+\delta)\right]\frac{p}{p-1}} \left(\mathbf{V}_{\gamma} g\right)^{\frac{p}{p-1}}(y) dy \right\}^{\frac{p-1}{p}} \\ &= \left\{ \int_{\mathbb{R}^{\mathbf{N}}} |y|^{\left[\alpha-(\gamma+\delta)\right]\frac{p}{p-1}} \left(\mathbf{V}_{\gamma} g\right)^{\left[\frac{p}{p-1}-\frac{q}{q-1}\right]}(y) \left(\mathbf{V}_{\gamma} g\right)^{\frac{q}{q-1}}(y) dy \right\}^{\frac{p-1}{p}}.\end{aligned} \tag{3. 37}$$

By using (3. 33) again, we have

$$\begin{aligned}|y|^{\left[\alpha-(\gamma+\delta)\right]\frac{p}{p-1}} \left(\mathbf{V}_{\gamma} g\right)^{\left[\frac{p}{p-1}-\frac{q}{q-1}\right]}(y) &\leq \mathfrak{C}_{\gamma\ q}\, |y|^{\left[\alpha-(\gamma+\delta)\right]\frac{p}{p-1}} |y|^{-\mathbf{N}\left(\frac{q-1}{q}\right)\left[\frac{p}{p-1}-\frac{q}{q-1}\right]} \left\| g \right\|_{\mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^{\mathbf{N}})}^{\frac{p}{p-1}-\frac{q}{q-1}} \\ &= \mathfrak{C}_{\gamma\ q}\, |y|^{\mathbf{N}\left[\frac{1}{p}-\frac{1}{q}\right]\frac{p}{p-1}-\mathbf{N}\left(\frac{q-1}{q}\right)\left[\frac{p}{p-1}-\frac{q}{q-1}\right]} \left\| g \right\|_{\mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^{\mathbf{N}})}^{\frac{p}{p-1}-\frac{q}{q-1}} \\ &= \mathfrak{C}_{\gamma\ q} \left\| g \right\|_{\mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^{\mathbf{N}})}^{\frac{p}{p-1}-\frac{q}{q-1}}.\end{aligned} \tag{3. 38}$$

From (3. 35)-(3. 38), together with the $\mathbf{L}^{\frac{q}{q-1}}$-estimate in (3. 30), we conclude

$$\left\| \mathbf{U}_2 f \right\|_{\mathbf{L}^q(\mathbb{R}^{\mathbf{N}})} \leq \left\| f \right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \qquad 1 \leq p < q < \infty. \tag{3. 39}$$

Recall $\mathbf{U}_3 f$ defined in (3. 10). We have

$$\begin{aligned}\mathbf{U}_3 f(x) &= \int_{\frac{1}{2}|x|<|y|<2|x|} f(y)\left(\frac{1}{|x|}\right)^{\gamma}\left(\frac{1}{|x-y|}\right)^{\mathbf{N}-\alpha}\left(\frac{1}{|y|}\right)^{\delta} dy \\ &\lesssim \int_{\mathbb{R}^{\mathbf{N}}} f(y)\left(\frac{1}{|x-y|}\right)^{\mathbf{N}-(\alpha-\gamma-\delta)} dy. \qquad (\gamma+\delta\geq 0)\end{aligned} \tag{3. 40}$$

Let $\frac{\alpha}{\mathbf{N}}=\frac{1}{p}-\frac{1}{q}+\frac{\gamma+\delta}{\mathbf{N}}, 1\leq p<q<\infty$. Define

$$\mathbf{I}_{\alpha-\gamma-\delta} f(x) = \int_{\mathbb{R}^{\mathbf{N}}} f(y)\left(\frac{1}{|x-y|}\right)^{\mathbf{N}-(\alpha-\gamma-\delta)} dy. \tag{3. 41}$$

For $p>1$, we have

$$\left\|\mathbf{I}_{\alpha-\gamma-\delta} f\right\|_{\mathbf{L}^q(\mathbb{R}^{\mathbf{N}})} \leq \mathfrak{C}_{p\ q} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}. \tag{3. 42}$$

This is a classical result due to Hardy, Littlewood and Sobolev [1]-[2].

We therefore conclude

$$\left\|\mathbf{U}_3 f\right\|_{\mathbf{L}^q(\mathbb{R}^{\mathbf{N}})} \leq \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \qquad p>1. \tag{3. 43}$$

For $p=1$, we assert

$$\begin{aligned}\mathbf{U}_3 f(x) &= \int_{\frac{1}{2}|x|<|y|<2|x|} f(y)\left(\frac{1}{|x|}\right)^{\gamma}\left(\frac{1}{|x-y|}\right)^{\mathbf{N}-\alpha}\left(\frac{1}{|y|}\right)^{\delta} dy \\ &\lesssim \int_{\frac{1}{2}|x|<|y|<2|x|} f(y)|y|^{-\varepsilon}\left(\frac{1}{|x|}\right)^{\gamma}\left(\frac{1}{|x-y|}\right)^{\mathbf{N}-\alpha}\left(\frac{1}{|y|}\right)^{\delta-\varepsilon} dy \\ &\lesssim \int_{\frac{1}{2}|x|<|y|<2|x|} f(y)|y|^{-\varepsilon}\left(\frac{1}{|x-y|}\right)^{\mathbf{N}-(\alpha-\gamma-\delta)-\varepsilon} dy, \qquad \varepsilon>0\end{aligned} \tag{3. 44}$$

proved that $\gamma+\delta>0$ and $\varepsilon>0$ is sufficiently small.

By using (3. 44) and applying Minkowski integral inequality, we have

$$\begin{aligned}\left\|\mathbf{U}_3 f\right\|_{\mathbf{L}^q(\mathbb{R}^n)} &\lesssim \left\{\int_{\mathbb{R}^{\mathbf{N}}}\left\{\int_{\frac{1}{2}|x|<|y|<2|x|} f(y)|y|^{-\varepsilon}\left(\frac{1}{|x-y|}\right)^{\mathbf{N}-(\alpha-\gamma-\delta)-\varepsilon} dy\right\}^q dx\right\}^{\frac{1}{q}} \\ &\leq \int_{\mathbb{R}^{\mathbf{N}}} f(y)|y|^{-\varepsilon}\left\{\int_{\frac{1}{2}|y|<|x|<2|y|}\left(\frac{1}{|x-y|}\right)^{q\left[\mathbf{N}-(\alpha-\gamma-\delta)-\varepsilon\right]} dx\right\}^{\frac{1}{q}} dy.\end{aligned} \tag{3. 45}$$

Note that $q[\mathbf{N}-(\alpha-\gamma-\delta)]=\mathbf{N}$. A direct computation shows

$$\int_{\frac{1}{2}|y|<|x|<2|y|}\left(\frac{1}{|x-y|}\right)^{q\left[\mathbf{N}-(\alpha-\gamma-\delta)-\varepsilon\right]} dx \lesssim |y|^{q\varepsilon}. \tag{3. 46}$$

By putting together (3. 45) and (3. 46), we obtain

$$\left\|\mathbf{U}_3 f\right\|_{\mathbf{L}^q(\mathbb{R}^{\mathbf{N}})} \leq \left\|f\right\|_{\mathbf{L}^1(\mathbb{R}^{\mathbf{N}})}. \tag{3. 47}$$

References

[1] G. H. Hardy and J. E. Littlewood, *Some Properties of Fractional Integrals*, Mathematische Zeitschrift **27**: no.1, 565-606, 1928.

[2] S. L. Sobolev, *On a Theorem of Functional Analysis*, Matematicheskii Sbornik **46**: 471-497, 1938.

[3] E. M. Stein and G. Weiss, *Fractional Integrals on n-Dimensional Euclidean Space*, Journal of Mathematics and Mechanics **7**: 503-514, 1958.

[4] E. M. Stein, *Singular Integrals and Differentiability Properties of Functions*, Princeton University Press, 1970.

[5] C. Fefferman and B. Muckenhoupt, *Two Nonequivalent Conditions for Weight Functions*, Proceeding of the American Mathematical Society **45**: 99-104, 1974.

[6] B. Muckenhoupt and R. L. Wheeden, *Weighted Norm Inequality for Fractional Integrals*, Transactions of the American Mathematical Society **192**: 261-274, 1974.

[7] E. T. Sawyer and R. L. Wheeden, *Weighted Inequalities for Fractional Integrals on Euclidean and Homogeneous Spaces*, American Journal of Mathematics **114**: no. 4, 813-874, 1992.

[8] P. Nápoli and T. Picon, *Stein-Weiss inequality in* $\mathbf{L}^1$*-norm for vector fields*, Proceedings of the American Mathematical Society **151**: no. 4, 1663-1679, 2023.

[9] C. Perez, *Two Weighted Norm Inequalities for Riesz Potentials and Uniform* $\mathbf{L}^p$*-Weighted Sobolev Inequalities*, Indiana University Mathematics Journal **39**: no.1, 31-44, 1990.

School of Mathematical Sciences, Zhejiang University, Hangzhou, 310058, China
email: sunchuhan@zju.edu.cn

School of Science, Westlake University, Hangzhou, 310030, China
email: wangzipeng@westlake.edu.cn